CPOR: Cumulative Protection for Online Reconciliation with Nodewise Error Limits

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Abstract

Forecast reconciliation can improve overall accuracy while worsening individual regions. We propose Cumulative Protection for Online Reconciliation (CPOR), which limits each node's cumulative root mean squared error (RMSE) relative to a causal reference from an agreed start. CPOR derives an admissible prediction interval from each node's remaining error allowance. Combining these intervals with aggregation constraints gives the exact one-step protection set for nonnegative observations with no known upper bound. Updating allowances using the errors of released forecasts maintains these limits after every observation. Two implementations combine candidate forecasts: Cov fits weights to recent forecast errors, and ML updates weights sequentially. Experiments on seven datasets use three forecasting backbones, each trained three times. We compare release rules using identical reconciled forecasts from Post-forecasting Identification and Revision (PIR) under the same protection limits. Full projection reduces mean-node RMSE relative to common shrinkage by 3.57% in window tests and 4.34% in continuous tests. The reductions relative to two-group shrinkage are 2.72% and 3.28%. All tested protected paths satisfy the cumulative limits. Complete-procedure comparisons favor Cov and ML over adapted ProCAST and Protected PIR on average. Average paired comparisons favor unprotected PIR in window tests and CPOR in continuous tests; PIR is more accurate with DUET in both. Protection can increase error at individual regions or system totals. Full projection retains more accuracy than the tested shrinkage rules when these cumulative limits are required.

Keywords: hierarchical forecasting; online learning; forecast reconciliation; cumulative error protection; forecast post-processing.

1 Introduction

Regional demand forecasts should add up to system demand, and store sales forecasts should agree with state and national totals. Forecasts that satisfy these aggregation relationships are called coherent [1]. Reconciliation adjusts separately generated forecasts using these known relationships [2].

Improved overall accuracy can conceal deterioration at individual nodes: a revision may improve the total while worsening regional forecasts. Regions responsible for inventory or scheduling may therefore require their cumulative errors to remain within a specified proportion of an agreed reference's error.

Minimum trace (MinT) reconciliation minimizes total reconciled error variance under unbiasedness [3]. Game-theoretically optimal reconciliation (GTOP) controls the worst-case increase in weighted total squared error [4]. ProCAST learns forecast revisions under coupled aggregation constraints [5]. These objectives do not impose a separate cumulative error limit at every node. Conservative online convex optimization constrains cumulative loss relative to a default strategy [6], with guarantees under bounded-loss and controlled-gradient assumptions. Here, forecast errors may be unbounded.

We study how to improve regional and system forecasts while maintaining aggregation consistency and a separate cumulative error limit for every observed node. A forecast revision service assessed over an agreed reporting period is one example: the reference rule and start time are fixed in advance, and each reference forecast uses only information available at prediction time. Two difficulties arise. The next observation is unknown when a forecast is released, and regional adjustments are linked through the aggregation constraints. Independent corrections can break coherence, while total-error optimization can overlook a node's limit.

We propose Cumulative Protection for Online Reconciliation (CPOR), which converts each node's remaining error allowance into an admissible interval. Combining these intervals with the aggregation constraints gives the exact one-step protection set for nonnegative outcomes with no known upper bound. After each observation, CPOR updates the allowances using the error of the forecast actually released. Cov chooses a combination of candidate forecasts at each node using their recent errors. ML updates these combination weights as new errors are observed. Both reconcile the combined target and apply the same protection operator.

The main contributions are as follows:

- We derive the exact one-step protection set for nonnegative outcomes with no known upper bound. The nodewise intervals and aggregation constraints give necessary and sufficient conditions for the next coherent forecast to preserve every node's limit for all admissible observations.
- We construct an online procedure that maintains these limits using actual released errors. The protection guarantee applies to any causal selection rule satisfying the set constraints; Cov and ML provide two concrete implementations.
- We evaluate both implementations on seven datasets using PatchTST, DLinear and DUET forecasts with three training runs. Window and continuous tests measure the accuracy of complete procedures and the effect of retaining unused allowance. We also give the same reconciled PIR forecasts to full projection and shrinkage rules under the same protection limits. Regional and system RMSEs show where forecast accuracy improves or deteriorates.

2 Related work

2.1 Forecast reconciliation and protection objectives

Classical reconciliation combines forecasts from different levels to satisfy aggregation constraints [2]. MinT chooses an unbiased reconciliation that minimizes the trace of the forecast-error covariance [3]. Nonnegative reconciliation adds the sign constraints needed for quantities such as demand and sales [7]. Robust reconciliation accounts for uncertainty in the estimated error covariance [8].

Projection geometry explains how reconciliation affects forecast loss and bias [9]. A complementary view treats reconciliation as combining direct forecasts with indirect forecasts derived from the aggregation constraints [10]. Li et al.'s recent preprint develops this view into a penalized combination framework for bottom-level series, whose revised forecasts are aggregated to recover the hierarchy [11]. These approaches characterize coherent adjustment and its accuracy properties. CPOR derives the additional restrictions imposed by each node's realized cumulative error limit.

GTOP provides a worst-case guarantee for weighted total squared error at a single forecast issue [4]. It projects the supplied forecasts onto the coherent outcomes permitted by a closed convex information set. CPOR imposes a separate cumulative constraint at every node and uses past released errors to determine the next permissible correction.

Other reconciliation objectives protect forecasts or economic outcomes. Immutable reconciliation keeps selected forecasts unchanged [12]. Value-oriented reconciliation uses Nash bargaining

to coordinate agents' economic gains [13]. CPOR allows forecasts at all nodes to change while limiting their cumulative prediction errors relative to a causal reference.

2.2 Learning to reconcile and revise forecasts

Learning-based methods adapt the forecaster, the reconciliation map or the forecast combination. Learning optimal projection jointly trains a neural forecaster and an oblique reconciliation layer [14]. The Flow-based Reconcile Transformer (FRT) combines an autoregressive transformer with a conditional normalizing flow to produce coherent probabilistic forecasts [15]. DUET models temporal heterogeneity and channel dependence through dual clustering [16]. ProCAST uses virtual underlying sequences in orthogonal and learned-oblique projections to enforce coupled aggregation constraints [5]. For online prediction, sequential expert aggregation can be followed by projection onto the coherent subspace [17]. CPOR can use these prediction rules to construct a target, then restricts the released forecast to satisfy every node's cumulative error limit.

Forecast post-processors revise predictions from a frozen backbone. The output branch of δ-Adapter learns residual corrections [18]. PIR identifies unreliable predictions and revises them using local covariates and retrieved historical instances [19]. COSA applies gated residual correction using recent observations [20]. Biswas et al. train a separate residual forecaster and subtract its scaled predictions from the base forecasts [21]. Under-Cali uses uncertainty-driven experts to calibrate inputs and outputs for irregularly sampled series [22]. Such corrections address forecast accuracy; CPOR determines which corrected targets can be published while preserving aggregation consistency and cumulative limits.

Calibration has a different objective. MCGrad improves probability calibration across subgroups [23]. AdaPI adjusts prediction interval widths using observed coverage [24]. CPOR intervals restrict the point forecast that may be released under a cumulative error limit.

2.3 Online learning with cumulative protection

Conservative online convex optimization constrains cumulative loss relative to a fixed default strategy [6]. Its Conservative Projection algorithm preserves this constraint while retaining the underlying learner's regret order under bounded-loss and gradient conditions. CPOR considers one squared-error constraint per node, coupled through the aggregation structure, with nonnegative observations of unbounded size. For this problem, CPOR derives the exact interval of allowed values at each node. A forecast must lie within every interval and satisfy the aggregation constraints. Updating allowances from released forecasts' errors then maintains the cumulative limits.

3 Problem formulation

We study online forecast reconciliation with a cumulative error constraint for each observed node. Table 1 summarizes the notation; quantities used only by the algorithm are defined in Section 4. Here $\mathbb{R}_+$ denotes the nonnegative real numbers. Unless stated otherwise, vector inequalities apply separately to each entry. The positive-part operator $[x]_+ = \max(x, 0)$ sets negative values to zero in the protection bounds. The symbols $\mathbf{1}$, I and $\|\cdot\|_2$ denote the all-ones vector, an identity matrix and the Euclidean norm. The symbol $\top$ denotes transpose.

A subscript i denotes a node, and j a candidate or underlying component as specified locally. The time indices s and t denote a historical forecast release and the current release, respectively. A time subscript alone denotes a vector across all nodes: for example, $p_t = (p_{t,1}, \ldots, p_{t,n})^\top$, where n is the number of observed nodes. The additional index i selects one scalar entry, so $p_{t,i}$ is the forecast for node i at time t. The same convention applies to f_t, b_t, y_t and the state vectors A_t, E_t. The states A_t, E_t are recorded before the corresponding release.

For a positive integer K, the simplex $\Delta_K = \{w \in \mathbb{R}_+^K : \sum_{j=1}^K w_j = 1\}$ contains nonnegative weights summing to one. Each implementation uses such weights to combine K raw experts at a node, before imposing the aggregation constraints.

Table 1: Notation for the problem and protection procedure.

Symbol	Meaning
$S, \mathcal{C}$	Fixed aggregation matrix and set of nonnegative forecasts consistent with it
f_t, b_t, p_t, y_t	Vectors across all n nodes at time t: base forecast, reference, released forecast and observation
$f_{t,i}, b_{t,i}, p_{t,i}, y_{t,i}$	Scalar entries for node i in these vectors
ϵ_i, R_i	Node i's relative RMSE allowance and squared factor $R_i = (1+\epsilon_i)^2$
A_t, E_t	Vectors of remaining error allowances and cumulative squared errors before release t
$A_{t,i}, E_{t,i}$	Scalar entries of these states for node i
$\Gamma_R(\beta, x)$	Scalar excess-error bound for reference value β, candidate value x and factor $R > 1$
$\mathcal{K}_t$	Set of admissible forecast vectors at time t
a_t, z_t, c_t	Raw combined target, coherent target and protected forecast
u	Nonnegative underlying values representing a coherent forecast
K, Δ_K	Number of raw experts and simplex of nodewise combination weights
v_i	Fixed positive pre-calibration variance used to scale adjustments

Aggregation structure. Let m be the number of underlying quantities. A fixed matrix $S \in \mathbb{R}_+^{n \times m}$ specifies how the underlying quantities contribute to each observed node: entry S_{ij} is the weight of component j in node i. Every row is nonzero, so each node depends on at least one component.[1] For nonnegative underlying values u, the vector $S\,u$ gives the corresponding node forecasts. The set of allowable forecasts is therefore

$$\mathcal{C} = \{S\,u : u \in \mathbb{R}_+^m\}. \tag{1}$$

Membership in $\mathcal{C}$ means that a forecast satisfies the aggregation and nonnegativity requirements; we call such a forecast *coherent*. For two regions with forecasts of 40 and 60, the aggregation is

$$\underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}}_{S} \underbrace{\begin{pmatrix} 40 \\ 60 \end{pmatrix}}_{u} = \underbrace{\begin{pmatrix} 100 \\ 40 \\ 60 \end{pmatrix}}_{S\,u} \quad \begin{matrix} \text{system total} \\ \text{region 1} \\ \text{region 2} \end{matrix} . \tag{2}$$

Multiplying the aggregation matrix S by the underlying forecast vector u gives the forecasts for the system total and the two regions, in that order. The first row adds the regional forecasts; the other rows retain each region's forecast. More generally, rows may describe overlapping aggregates of observed or latent components. A nonnegative vector u satisfying $p = S\,u$ represents the forecast through its underlying components.

The set $\mathcal{C}$ is convex: a weighted average of its forecasts remains in $\mathcal{C}$ when the weights are nonnegative and sum to one. This structure permits convex projection onto the aggregation constraints and contraction of a coherent forecast toward the coherent reference.

We assume that observations $y_t \in \mathbb{R}_+^n$ are nonnegative, without requiring a known upper bound. When the observed series obey the same aggregation relationships, we additionally require $y_t \in \mathcal{C}$.

[1] A zero row fixes $b_{t,i} = p_{t,i} = 0$, whose excess loss is $(1 - R_i)y_{t,i}^2 \le 0$. Such fixed-zero coordinates can be omitted from optimization and restored as zero forecasts.

Information and reference forecasts. At time t, the forecaster receives a base prediction $f_t \in \mathbb{R}^n$ and publishes a reconciled prediction $p_t \in \mathcal{C}$. Reconciliation is the adjustment from the supplied prediction to a coherent one. A reference rule produces a coherent forecast $b_t \in \mathcal{C}$ for comparing errors. The rule is fixed before the evaluation period begins, but its forecast can change as new information becomes available. Both b_t and p_t may use the current base prediction, past observations and covariates available at issue t; neither may use y_t or later observations. We assume y_t is observed after publishing p_t and before issue $t+1$. A past forecast can be used for error-based updates once its observation has returned. We use *protection reference* or *anchor* for b_t, and *comparator* for a method used to evaluate accuracy.

Nodewise cumulative protection. Fix a start time $t = 1$ and a relative RMSE allowance $\epsilon_i > 0$ for each node i. For example, $\epsilon_i = 0.01$ permits RMSE at most 1% above the reference RMSE. The reference rule and allowances remain fixed throughout this period. The factor $R_i = (1+\epsilon_i)^2$ converts this RMSE limit to a squared-error limit. At every elapsed time T, require

$$\sum_{t=1}^{T}(p_{t,i} - y_{t,i})^2 \le R_i \sum_{t=1}^{T}(b_{t,i} - y_{t,i})^2, \qquad i = 1, \dots, n, \quad T \ge 1. \tag{3}$$

The left side is the cumulative squared error of the released forecasts; the right side is the permitted error relative to the reference. Dividing both sides by T and taking square roots gives the RMSE limit. A *prefix* $1{:}T$ consists of all observations from the fixed start through time T, so the requirement holds after every new observation. The guarantee applies separately to each node over prefixes from the fixed start; individual-date and rolling-window errors can exceed this limit.

Prediction objective. For the released sequence $p = (p_1, \dots, p_T)$, we measure joint forecast accuracy by mean-node RMSE,

$$M_T(p) = \frac{1}{n}\sum_{i=1}^{n}\sqrt{\frac{1}{T}\sum_{t=1}^{T}(p_{t,i} - y_{t,i})^2}. \tag{4}$$

The metric averages nodewise RMSE in the original data units with equal node weights; larger-scale errors therefore contribute more. We seek a causal policy using information available at release to achieve low $M_T(p)$, while maintaining $p_t \in \mathcal{C}$ and Eq. (3) at every prefix. Publishing the reference b_t is always feasible; the aim is to improve accuracy while retaining the nodewise limits.

4 The CPOR protection framework

CPOR first determines which forecasts satisfy the constraints in Section 3, then selects a forecast from this set. Figure 1 shows the sequence: construct the reference and candidate forecasts, combine them with Cov or ML, reconcile and protect the target, and update the remaining allowances after feedback. Table 2 summarizes the two implementations. They use the same reference rule and error tolerance, but each updates its own allowances using its released forecasts.

Table 2: Two implementations of the CPOR framework.

Implementation	Prediction selection	Weight structure	Protected targets
CPOR-Cov	Error-second-moment combination	Nine weights per node	One combined target
CPOR-ML	Adapted ML-Pol combination	Nine weights per node	One combined target

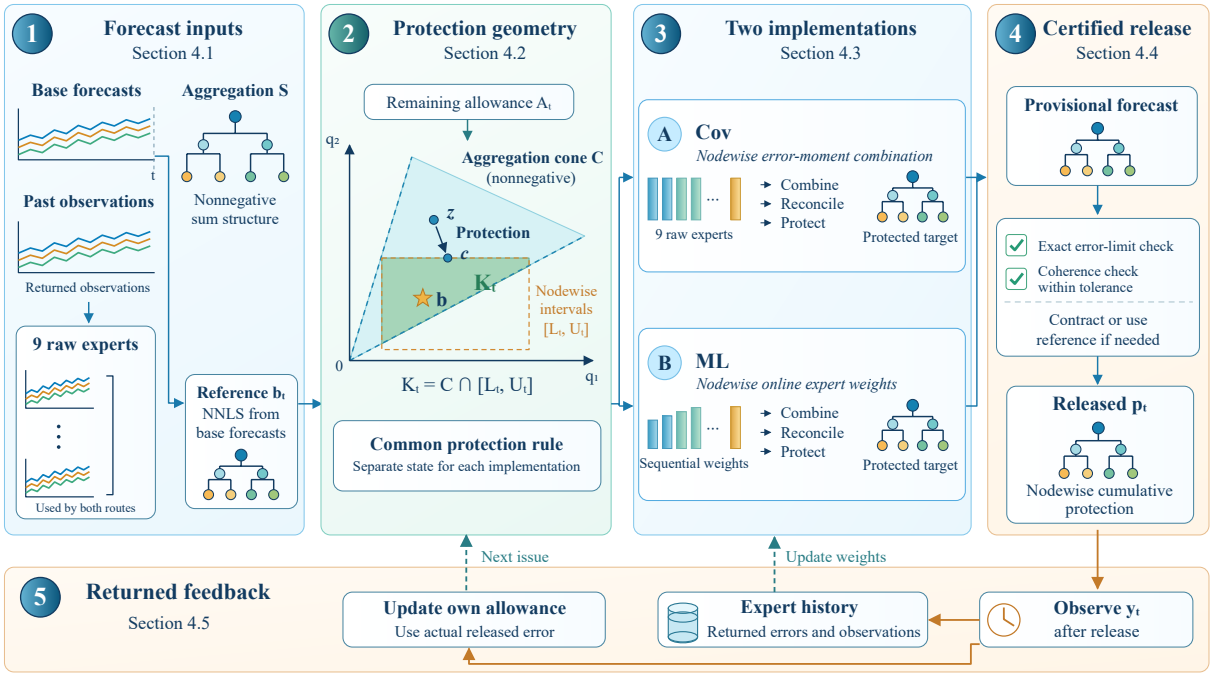


Figure 1: CPOR with two prediction-selection implementations. Cov and ML each combine raw experts, reconcile the target and apply the protection rule. Each maintains its own remaining allowances. Dashed arrows carry returned information to the next issue.

4.1 Reference and candidate construction

We first construct the reference used to measure permitted error. Given the supplied forecast f_t, we find its nearest nonnegative coherent forecast by nonnegative least squares (NNLS):

$$u_t^b \in \arg\min_{u \geq 0} \tfrac{1}{2} \|S\,u - f_t\|_2^2, \qquad b_t = S\,u_t^b. \tag{5}$$

Here $u_t^b \in \mathbb{R}_+^m$ represents the underlying quantities for the reference; the superscript b identifies its role. The rule stays fixed; its output changes with the supplied base forecast. Eq. (5) is also the mathematical problem solved by ProCAST's orthogonal branch; it is separate from the learned-oblique comparator used below. NNLS provides a coherent reference without estimating an error covariance matrix.

Both implementations start from the same nine forecasting rules, called raw experts. Each uses only information available before release: the supplied forecast, four bias-corrected forecasts, persistence, a recent-observation mean, a lagged observation and ridge error correction. The four bias corrections subtract mean base-forecast errors estimated over different history lengths, balancing responsiveness to recent changes against smoothing of individual errors. The observation-based rules supply the latest, smoothed and lagged levels; ridge regression estimates error from forecast and historical features. Write $a_t^{(j)} \in \mathbb{R}^n$ for raw expert j, with $K = 9$ experts.

For any raw target a_t, variance-weighted nonnegative reconciliation gives

$$z_t = \arg\min_{q \in \mathcal{C}} \frac{1}{2} \sum_{i=1}^{n} (q_i - a_{t,i})^2 / v_i, \tag{6}$$

where $v_i > 0$ is a fixed pre-calibration observation variance with a positive floor. Cov and ML each combine the raw experts into one target, then use this operation to make it coherent. Bias and ridge corrections use past errors of the base forecast. Cov and ML weights use past errors of the raw experts. The remaining allowance is updated separately using the error of the forecast actually released.

4.2 Exact nodewise protection

Remaining allowances and worst-case excess error. Before issue t, let $E_{t,i}$ be the cumulative squared error of actual releases and $A_{t,i}$ the remaining error allowance at node i:

$$E_{t,i} = \sum_{s<t} (p_{s,i} - y_{s,i})^2,$$
$$A_{t,i} = R_i \sum_{s<t} (b_{s,i} - y_{s,i})^2 - E_{t,i}. \tag{7}$$

The factor $R_i = (1+\epsilon_i)^2$ fixes the permitted cumulative squared error relative to the reference.

After the next observation, every node's remaining allowance must still be nonnegative. To see what this requires, consider one node with reference value $\beta \geq 0$, candidate forecast $x \geq 0$ and squared-error factor $R > 1$. If the next observation is $y \geq 0$, the allowance decreases by

$$(x-y)^2 - R(\beta - y)^2. \tag{8}$$

This is the candidate's squared error minus the permitted contribution from the reference error. A positive value consumes allowance; a negative value adds allowance. Protection therefore requires this excess error to be no greater than the allowance available before release.

Because y is unknown at release time, we control the largest excess error over all nonnegative observations. Define

$$\Gamma_R(\beta, x) := \sup_{y \geq 0} \{(x-y)^2 - R(\beta - y)^2\}. \tag{9}$$

Thus Γ_R is the largest amount of allowance the candidate could consume at the next observation. At time t, a candidate vector $q = (q_1, \ldots, q_n)^\top$ uses $(\beta, x, R) = (b_{t,i}, q_i, R_i)$ at node i. Combining all nodewise checks with aggregation consistency gives

$$\mathcal{K}_t = \{q \in \mathcal{C} : \Gamma_{R_i}(b_{t,i}, q_i) \leq A_{t,i},\ i = 1, \ldots, n\}. \tag{10}$$

The supremum in Eq. (9) ranges over infinitely many possible observations. The next theorem gives a closed-form expression that makes the check computable. It also establishes that $\mathcal{K}_t$ contains exactly the coherent forecasts that preserve every node's remaining allowance after the next observation, and that this set is nonempty and convex.

Theorem 1 (Exact one-step protection geometry)**.** *For scalar $\beta, x \geq 0$ and $R > 1$, the worst-case excess error in Eq.* (9) *is*

$$\Gamma_R(\beta, x) = x^2 - R\beta^2 + \frac{[R\beta - x]_+^2}{R-1}. \tag{11}$$

Let $S \in \mathbb{R}_+^{n \times m}$ have no zero row, $b_t \in \mathcal{C}$, and $A_t \geq 0$. For either outcome domain $\mathcal{Y} = \mathbb{R}_+^n$ or $\mathcal{Y} = \mathcal{C}$, $\mathcal{K}_t$ is nonempty and convex. A coherent forecast belongs to $\mathcal{K}_t$ if and only if it leaves every remaining allowance nonnegative for every $y_t \in \mathcal{Y}$.

Proof. The excess loss equals $-(R-1)y^2 + 2(R\beta - x)y + x^2 - R\beta^2$. Its maximizer on the nonnegative half-line is $y^* = \max\{0, (R\beta - x)/(R-1)\}$. Substituting this value gives Eq. (11). For each fixed y, the excess loss has second derivative 2 in x. Its pointwise supremum over $y \geq 0$ is finite by $R > 1$ and therefore convex in x. Also $\Gamma_R(\beta, \beta) = 0$, so $b_t \in \mathcal{K}_t$. Intersecting the convex nodewise sublevel sets with the convex cone $\mathcal{C}$ proves convexity of $\mathcal{K}_t$; the anchor proves nonemptiness. For any candidate vector $q \in \mathcal{K}_t$, the excess loss at each node is at most its available allowance, so updating the state cannot make that allowance negative. Conversely, violating any envelope constraint permits a nonnegative outcome at its scalar maximizer that makes that node's remaining error allowance negative. For the coherent domain, it suffices to construct a coherent observation that reaches this worst value at the violated node. For that row i, let y_i^* be the scalar maximizer with $\beta = b_{t,i}$ and $x = q_i$. Choose j with $S_{ij} > 0$, set

$u_j = y_i^*/S_{ij}$ and all other components to zero. The coherent observation $\widetilde{y} = S\,u \in \mathcal{C}$ has value y_i^* at node i, so that node's allowance becomes negative. A violation at this single node suffices to prove necessity. For example, let the rows of S be $(1,1,0)$, $(0,1,1)$ and $(1,1,1)$. They overlap in their second component. For any $a \geq 0$, $u = (0,a,0)^\top$ gives $S\,u = (a,a,a)^\top$, attaining $y_1^* = a$. Nonnegativity permits the zero components used in this construction. □

Additional restrictions, such as upper bounds or fixed positive component totals, would change the outcome domain. For a smaller domain than $\mathcal{C}$, the present envelopes give a sufficient protection condition; the exact set depends on the added restrictions.

The nonnegative outcome domain matters when $x > R\beta$. If y were allowed to range over all real values, the supremum would be $R(x-\beta)^2/(R-1)$. For $x \leq R\beta$ its maximizing outcome is already nonnegative, so the envelopes agree. When $x > R\beta$, the excess loss decreases from $y = 0$, so its maximum is $x^2 - R\beta^2$: predicting a large positive value is most costly when the outcome is zero. The nonnegative-domain maximum is smaller by $(x - R\beta)^2/(R-1)$. The resulting upper bound on the forecast can therefore be larger, even without an upper bound on the observation. For example, $R = 2$, $\beta = 1$ and $A = 4$ give upper bounds $\sqrt{6}$ and $1 + \sqrt{2}$ for the nonnegative and all-real envelopes, respectively.

Admissible intervals and projection. Eq. (11) computes the worst-case allowance consumption of a given candidate value x. To construct a forecast, we instead solve for all nonnegative values of x allowed by a given reference β and remaining allowance $A \geq 0$. These values satisfy $\Gamma_R(\beta, x) \leq A$. To solve this inequality, we expand the positive-part term in Eq. (11). With $\eta = R - 1 > 0$, this gives the equivalent piecewise expression

$$\Gamma_R(\beta, x) = \begin{cases} R(x-\beta)^2/\eta, & x \leq R\beta, \\ x^2 - R\beta^2, & x > R\beta. \end{cases} \tag{12}$$

In the first branch of Eq. (12), the allowance constraint gives $|x - \beta| \leq \sqrt{A\eta/R}$; in the second, it gives $x \leq \sqrt{A + R\beta^2}$. Combining these solutions with their branch conditions and $x \geq 0$ yields the full interval of admissible values. Denote its lower and upper endpoints by

$$\begin{aligned} L(\beta, A) &= \max\{0, \beta - \sqrt{A\eta/R}\}, \\ U(\beta, A) &= \begin{cases} \beta + \sqrt{A\eta/R}, & A \leq R\eta\beta^2, \\ \sqrt{A + R\beta^2}, & A > R\eta\beta^2. \end{cases} \end{aligned} \tag{13}$$

The interval endpoints also depend on the fixed factor R, which is omitted from their notation. For $x \geq 0$, the test $\Gamma_R(\beta, x) \leq A$ is exactly equivalent to $L(\beta, A) \leq x \leq U(\beta, A)$. Thus Eq. (13) converts the worst-case error constraint into explicit bounds that can be imposed when adjusting candidate forecasts. The intervals restrict individual values; their intersection with $\mathcal{C}$ enforces the aggregation relations jointly. The upper-bound formula switches branches when the admissible endpoint reaches $x = R\beta$, where $\Gamma_R(\beta, R\beta) = R\eta\beta^2$; both expressions for U agree at this threshold.

The bounds also describe the limiting cases. With $A = 0$, both endpoints equal β, so only the reference value is admissible. More generally, Eq. (12) gives $\Gamma_R(\beta, x) \geq (x - \beta)^2$. Starting from $A_1 = 0$ therefore gives $p_1 = b_1$ and $A_{2,i} = (R_i - 1)(b_{1,i} - y_{1,i})^2$: following the anchor earns initial allowance when its error is nonzero. When $\beta = 0$, the interval is $[0, \sqrt{A}]$. With nonnegative observations, fixed S and feedback before the next release, the cumulative guarantee holds even when outcomes are unbounded, their distribution changes, or the forecasting model is misspecified. The error limit remains relative to the reference, whose own error can be large.

We protect the coherent target $z_t \in \mathcal{C}$ in two steps. The target uses only information available at time t. Although it already satisfies the aggregation constraints, it may exceed the remaining error allowances.

First, determine the feasible set. Evaluate Eq. (13) with $(\beta, A, R) = (b_{t,i}, A_{t,i}, R_i)$ at each node, and collect the bounds into vectors L_t and U_t. The admissible set is then

$$\mathcal{K}_t = \{q \in \mathcal{C} : L_t \leq q \leq U_t\}. \tag{14}$$

Both requirements matter: each entry must lie within its protection interval, and the whole vector must satisfy the aggregation relationships. The common-rule panel in Figure 1 shows this construction. Cov and ML use their own remaining allowances, so their admissible intervals can differ even with the same reference.

Second, select the closest feasible forecast to the target. To retain the target's information while enforcing protection, we choose a weighted squared distance as the adjustment objective. For fixed node scales $v_i > 0$, this gives

$$c_t \in \operatorname*{arg\,min}_{q \in \mathcal{K}_t} \tfrac{1}{2} \sum_i (q_i - z_{t,i})^2 / v_i. \tag{15}$$

Here q is the vector being optimized and c_t is the resulting protected candidate. The factors $1/v_i$ account for differences in node scale. With a fixed target and the same allowances, full projection minimizes this weighted distance over all feasible forecasts. A feasible forecast obtained by shrinking toward the reference cannot have a smaller target distance. Either forecast may still have a smaller error after the next observation. If the target already belongs to $\mathcal{K}_t$, the projection leaves it unchanged.

Writing $q = S\,u$ makes Eq. (15) a quadratic program (QP) with $u \geq 0$ and $L_t \leq S\,u \leq U_t$. The joint optimization preserves aggregation consistency. The set $\mathcal{K}_t$ is the full one-step robust protection set; each implementation selects one point in it by projecting its target.

Accuracy cost of protection. The intervals can force a forecast away from the value that is later observed. Write $I_{t,i} = [L(b_{t,i}, A_{t,i}), U(b_{t,i}, A_{t,i})]$, using R_i in each coordinate. For any realized y_t and protected p_t,

$$\sum_i (p_{t,i} - y_{t,i})^2 \geq \sum_i \operatorname{dist}(y_{t,i}, I_{t,i})^2. \tag{16}$$

Here $\operatorname{dist}(y, I) = \inf_{x \in I} |y - x|$. This bound applies to the current allowance state. Requiring aggregation consistency can increase this minimum error. Later forecast errors change the allowance and therefore the intervals. At zero remaining allowance, only b_t is allowed, even if an unprotected coherent forecast equals y_t. This explains how protection can increase error. A weighted least-squares (WLS), MinT or other reconciled forecast satisfies the one-step contract exactly when it lies in $\mathcal{K}_t$. The resulting accuracy cost depends jointly on the reference, realized errors, remaining allowance and aggregation structure.

4.3 Prediction selection: Cov and ML

Cov estimates relationships among recent expert errors, whereas ML updates weights from sequential relative gains. These are alternative ways to form a target; both use the protection operator in Section 4.2 and the actual-error update in Section 4.5.

4.3.1 CPOR-Cov: second-moment combination

Cov chooses a separate combination at each node using recent expert errors. A rolling history lets the error-moment estimate respond to changes in expert performance. Let H_t index this history before issue t, with its maximum length fixed before scoring. If H_t is empty, Cov uses uniform weights. Otherwise, define the normalized raw-expert error vector $r_{s,i} \in \mathbb{R}^K$ by $(r_{s,i})_j = (a_{s,i}^{(j)} - y_{s,i})/\sqrt{v_i}$. Its second-moment estimate and regularized matrix are

$$G_{t,i} = \frac{1}{|H_t|} \sum_{s \in H_t} r_{s,i} r_{s,i}^\top, \qquad Q_{t,i} = (1-\rho) G_{t,i} + \rho \operatorname{diag}(G_{t,i}) + \tau \gamma_{t,i} I_K, \tag{17}$$

where $\rho \in [0,1]$ is the diagonal shrinkage weight, $\tau > 0$ the ridge coefficient, I_K the identity, and $\gamma_{t,i} = \max\{\operatorname{tr}(G_{t,i})/K, \gamma_{\min}\}$ a positive scale with a fixed floor $\gamma_{\min} > 0$. Diagonal shrinkage reduces the contribution of cross-expert moments, while the positive ridge term keeps the fitting matrix positive definite. Uncentred moments include systematic bias as well as variability. Cov estimates error relationships across experts at each node; MinT estimates error covariance across nodes.

Cov minimizes a regularized estimate of the combined forecast's squared error. The resulting weights and raw target are

$$w_{t,i}^{\mathrm{Cov}} \in \arg\min_{w \in \Delta_K} w^\top Q_{t,i} w, \qquad a_{t,i}^{\mathrm{Cov}} = \sum_{j=1}^{K} w_{t,ij}^{\mathrm{Cov}} a_{t,i}^{(j)}. \tag{18}$$

Here $w_{t,ij}^{\mathrm{Cov}}$ is component j of the node-specific weight vector. The combined target is reconciled by Eq. (6), protected by Eq. (15) and released after certification.

4.3.2 CPOR-ML: online expert combination

ML updates the expert weights after each observation using an adapted ML-Pol rule [17]. At each node, let $h_{t,ij} = a_{t,i}^{(j)}/\sqrt{v_i}$ be the normalized expert forecast and let $\omega_{t,i} \in \Delta_K$ be the current weights. The normalized mixture is $m_{t,i} = \sum_j \omega_{t,ij} h_{t,ij}$. After y_t returns, define $\nu_{t,i} = y_{t,i}/\sqrt{v_i}$ and the linearized relative gain

$$g_{t,ij} = 2(m_{t,i} - \nu_{t,i})(m_{t,i} - h_{t,ij}), \qquad B_{t+1,ij} = B_{t,ij} + g_{t,ij}, \quad V_{t+1,ij} = V_{t,ij} + g_{t,ij}^2. \tag{19}$$

A positive gain favors moving toward that expert in a linear approximation of the mixture's squared error. The state B accumulates these gains, and V accumulates their squared magnitudes. Both start at zero. The next weights are

$$d_{t+1,ij} = \frac{\max\{B_{t+1,ij}, 0\}}{1 + V_{t+1,ij}}, \qquad \omega_{t+1,ij} = \frac{d_{t+1,ij}}{\sum_{k=1}^{K} d_{t+1,ik}}, \tag{20}$$

Negative accumulated gains give zero unnormalized scores. If all scores are zero, use uniform weights; otherwise normalize as above. Gains continue to accumulate even for zero-weight experts, so a positive B restores a positive score. Thus weights respond to experts' performance relative to the current combination. The raw target $a_{t,i}^{\mathrm{ML}} = \sqrt{v_i}\, m_{t,i}$ is reconciled by Eq. (6) and protected by Eq. (15). Node-specific weights can break aggregation consistency before this reconciliation. The common protection operator maintains the cumulative guarantee.

4.4 Certified forecast release

Before publication, the computed forecast must pass the original nodewise envelope tests in exact rational arithmetic and aggregation checks within numerical tolerance. Numerical safeguards narrow the projection intervals slightly toward the reference and, if necessary, contract the forecast and its underlying representation toward it. The accepted forecast is p_t. The release guard may return the reference; exhausted projection attempts stop the trajectory. Only accepted forecasts are released.

4.5 Feedback updates and cumulative protection

After observing y_t, we compute the released forecast's squared error and update E_t and A_t:

$$E_{t+1,i} = E_{t,i} + (p_{t,i} - y_{t,i})^2, \qquad A_{t+1,i} = A_{t,i} + R_i(b_{t,i} - y_{t,i})^2 - (p_{t,i} - y_{t,i})^2. \tag{21}$$

This update uses the feedback order in Section 3. Delayed or concurrent feedback requires reserving allowance for unsettled releases before further publication. Theorem 1 protects one release. The following corollary shows why repeating the rule protects the entire sequence: the update in Eq. (21) carries the remaining allowance forward using the error of the forecast actually released.

Corollary 2 (Nodewise prefix protection)**.** *Suppose the outcome domain and aggregation matrix satisfy Theorem 1. If $A_1 = 0$ and each actual publication belongs to $\mathcal{K}_t$, then Eq.* (3) *holds at every node and every prefix.*

Proof. Starting from $A_1 = 0$, Theorem 1 ensures that each release consumes no more than the available allowance. Eq. (21) therefore gives $A_{t+1,i} \geq 0$ at every step by induction. Rearranging Eq. (7) at time $T+1$ yields the cumulative error bound in Eq. (3) for every node and every prefix $1{:}T$. □

The guarantee applies to any causal reference in $\mathcal{C}$. For stored values, let $A^*_{t,i}$ be the exact remaining allowance and $A^{\text{cert}}_{t,i}$ the allowance used for certification. The release check enforces

$$\Gamma_{R_i}(b_{t,i}, p_{t,i}) \leq A^{\text{cert}}_{t,i} = A^*_{t,i}. \tag{22}$$

Since realized excess loss cannot exceed this envelope, updating the exact allowance by Eq. (21) preserves $A^*_{t+1,i} \geq 0$. Aggregation consistency is checked separately within numerical tolerance.

After feedback, the raw experts' error histories are extended. Cov updates its second-moment estimate; ML updates the gains in Eq. (19). These prediction-selection states use expert errors, whereas the remaining allowance uses the error of the forecast actually released.

Initialization. At the start of scoring, live cumulative errors E_t and remaining allowances A_t are reset to zero. Calibration observations and expert histories are retained for target generation; the new protection period accumulates allowance only from scored releases.

Algorithm 1 summarizes the procedure. The protection QP has m underlying variables, n interval constraints on $S\,u$ and m nonnegativity constraints.

Algorithm 1 CPOR with Cov or ML selection

Inputs: aggregation matrix S, reference rule, tolerances ϵ_i, scaling variances v_i, and returned expert history.

Initialize: reset protection states $E_1 = A_1 = 0$; retain calibration expert histories and Cov/ML selection states (including ML B, V).

For each release t:

1. Form the raw experts and the causal reference $b_t = Su^b_t$. Combine the experts into target a_t using Cov or ML, then reconcile it by Eq. (6) to obtain the coherent target z_t.
2. Compute the admissible intervals from b_t and A_t. Solve Eq. (15) for the protected candidate c_t and its underlying representation. Halt if the projection attempts are exhausted.
3. Apply the release checks in Section 4.4, including repair or contraction when required. The guard may return the reference. Store and publish p_t only after the envelope and coherence checks pass.
4. Receive y_t before the next release. Update E_{t+1}, A_{t+1} by Eq. (21) using p_t, and extend the expert histories. Update the Cov statistics or ML weights from the returned expert errors.

5 Experimental setup

5.1 Datasets

We evaluate seven datasets. Table 3 summarizes their aggregation dimensions, sampling intervals and units. RH and ECommerce use the data and aggregation matrices distributed with ProCAST [25]; Traffic, Wiki2 and Labour come from the hierarchical-data archive [26]. For M5 [27], we use a store–category subset aggregated to store, state and overall totals; "M5" refers to this subset throughout.

Traffic, Wiki2, Labour, M5 and AEMO have hierarchical aggregation structures. RH and ECommerce contain overlapping aggregates of latent components. RH includes a total over latent components; ECommerce has no observed overall total. All seven aggregation matrices are nonnegative and have no zero rows.

The Australian Energy Market Operator (AEMO) provides half-hourly estimated underlying electricity demand for New South Wales (NSW), Queensland (QLD), South Australia (SA), Tasmania (TAS) and Victoria (VIC). Underlying demand is operational demand plus rooftop photovoltaic (PV) generation [28]. The PV estimate scales measured output to represent regional rooftop generation [29]. The five regional series and their sum form six observed nodes, with $S = [\mathbf{1}^\top; I_5]$. September–October 2026 provides 2,928 complete half-hourly observations per region, aligned by interval end in Australian Eastern Standard Time (AEST, UTC+10). The resulting underlying-demand series are nonnegative; all regions and observations are retained without clipping.

Table 3: Datasets and aggregation dimensions.

Dataset	Observed n	Underlying m	Rank(S)	Interval	Unit
RH	7	31	7	Daily	Not specified
ECommerce	14	112	14	Business day	Items
Traffic	207	200	200	Daily	Dimensionless
Wiki2	199	150	150	Daily	Page views
Labour	57	32	32	Monthly	Thousand persons
M5	44	30	30	Daily	Items
AEMO	6	5	5	Half-hourly	MW

Items are units sold; MW denotes megawatts. Traffic contains dimensionless aggregated occupancy ratios. RH is a constructed exchange-rate benchmark without a specified physical unit.

5.2 Forecasting and evaluation protocols

We standardize evaluation windows, input length and forecast horizon [30]. PatchTST [31], DLinear [32] and DUET provide the base forecasts. Each trained model receives 24 recent observations and predicts a block of 12 values. These base forecasts remain fixed within the block. Postprocessors revise them sequentially: before release $h \in \{1, \ldots, 12\}$, they can use the $h-1$ observations already returned within that block, together with earlier history. All methods evaluated on a given stream receive identical base forecasts, target dates and feedback.

Table 4 gives two consecutive 72-observation windows for each dataset. Their calendar durations depend on the sampling interval. A configuration is a dataset–backbone–window combination: seven datasets, three backbones and two windows give 42 configurations. Training each configuration three times with independent initializations gives 126 base-forecast sequences. Validation mean absolute error (MAE) selects the model used to generate each sequence.

Window protocol. Each window uses 36 preceding calibration observations and starts scoring with zero remaining allowance. Variance weights are estimated before that calibration window. Calibration records initialize Cov, ML and the fixed expert control. Each protected rule then updates its own allowance from the errors of its actual releases.

Table 4: Early and late scoring windows.

Dataset	Early window	Late window
RH	2023-08-10–2023-10-20	2023-10-21–2023-12-31
ECommerce	2011-05-24–2011-08-31	2011-09-01–2011-12-09
Traffic	2008-08-10–2008-10-20	2008-10-21–2008-12-31
Wiki2	2016-08-10–2016-10-20	2016-10-21–2016-12-31
Labour	2008-12-01–2014-11-01	2014-12-01–2020-11-01
M5	2015-12-31–2016-03-11	2016-03-12–2016-05-22
AEMO	2026-10-29 00:30– 2026-10-30 12:00	2026-10-30 12:30– 2026-11-01 00:00

Dates are inclusive; AEMO uses fixed AEST (UTC+10) and interval-end labels. RH dates are synthetic sequence indices.

Continuous protocol. The 144 base forecasts from both windows are evaluated consecutively, starting with the early window's 36 calibration records. Variance weights, Cov/ML histories and the calibration-selected expert persist throughout. The continuous rule retains its allowance across all 144 releases. A control rule resets only the allowance before release 73. The two rules receive the same candidates and references; each updates its own allowance from the forecasts it releases. The late calibration observations fall within the first scored half and are not added again. Any backbone or adapter updates use only previously returned observations and are identical in the two paths.

Calibration forecasts are generated using only earlier observations. Historical forecast–observation pairs provide the support data used to fit the postprocessors. Model fitting and validation precede the forecasts they evaluate; the calibration and scoring boundaries are shared across methods on each base-forecast stream.

Longer continuous evaluation. We additionally evaluate AEMO over all 1,488 October half-hours to assess protection over a longer trajectory than the 144-observation tests. For every backbone and training run, each protected method starts with zero allowance and retains it throughout the month. The feedback schedule assumes that each finalized demand observation, including its PV component, is available before the next release. Supplementary Note S5 reports the whole-month comparisons and their evaluation scope.

RH, ECommerce, Traffic, Wiki2 and Labour were used during development. The expanded comparisons, training repetitions, and unified window and continuous tests were added after the initial development study.

5.3 Comparison methods

External postprocessors. The adapted ProCAST comparator uses its learned-oblique branch. Each update assigns total weight 0.5 to up to 72 preceding forecast–observation pairs and weight 0.5 to the current raw forecast treated as its own target. NNLS solves the orthogonal branch's optimization problem. Ada-Y and PIR are fitted on historical support pairs and selected using calibration data. Their fitted parameters remain fixed within each scoring window. Ada-Y refreshes its forecast correction; PIR uses the current base forecast, available observation context and calendar inputs. Each adapter is fitted once per base-forecast stream.

Matched publication rules. PIR denotes the postprocessor's direct output, without an added coherence or protection step. Coherent PIR applies variance-weighted nonnegative reconciliation to that output. Protected PIR then applies CPOR full projection to Coherent PIR. Cov and ML use the same reconciliation and protection steps on their own combined targets. Common shrinkage takes the largest permissible step from the NNLS reference toward the coherent candidate. Two-group shrinkage fixes two nearly equal-sized groups of underlying components using only the row supports and column order of S. It jointly chooses two step sizes to minimize the same

weighted target distance as full projection, subject to every node's interval. Full projection uses the complete protection set. The rules receive the same candidate, reference, variance weights, tolerance and feedback. Each updates its own allowance from its released errors. Comparing a protected rule with direct release of the coherent candidate measures its accuracy cost.

Protected controls. WLS uses calibration error variances. MinT uses covariance shrinkage on the same 36 calibration residuals [33], while rolling MinT uses the 36 most recent returned residuals. A feedback-matched MinT control instead uses the Cov target and its residual covariance. Each reconciled target receives CPOR protection under its own allowance. Protected CalBest keeps the expert with lowest calibration mean-node RMSE. ProCAST and Ada-Y targets are also evaluated with the same protection operator. These controls compare target construction and selection under a common error contract.

Supplementary comparisons with COSA, robust reconciliation, residual correction, the MC-Grad regression extension and Li et al.'s preprint use the earlier six-dataset, two-backbone streams. Their respective evaluation scopes are specified in Supplementary Note S8.

5.4 Parameters and computation

Table 5 lists the shared settings. The NNLS reference rule, allowance and hyperparameters are fixed before scoring.

Table 5: Shared experimental settings.

Parameter	Setting
Input / forecast horizon	24 / 12 observation steps
Training repetitions	Three per backbone
Calibration / window score	36 / 72 observations
Continuous scored length	144 observations
Midpoint-reset control	Allowance reset before release 73
AEMO whole-month score	1,488 half-hours; no reset
Cov moment history	Up to 28 returned observations
Nodewise allowance ϵ_i	0.005
Initial protection allowance	Zero at the agreed scoring start

CPOR publication uses one CPU thread on an AMD Ryzen 9 8945HX. Neural training uses an NVIDIA RTX 5070 Laptop GPU.

5.5 Evaluation metrics

1. **Mean-node RMSE.** We use Eq. (4), with lower values indicating smaller error. Window summaries first average early and late RMSE within each dataset, backbone and training run. Continuous RMSE is computed directly over all 144 observations; monthly RMSE uses all 1,488 AEMO observations.

2. **Paired RMSE reduction.** Let M_{impl} and M_{comp} denote the implementation's and comparator's mean-node RMSEs under the same protocol. The reduction is $100(1 - M_{\text{impl}}/M_{\text{comp}})$, computed within each matched run. We average these reductions over the three runs and 21 dataset–backbone groups with equal weights. Protection cost is the percentage increase in RMSE relative to the same unprotected coherent candidate.

3. **Aggregation-level RMSE.** We separately evaluate observed system totals, Labour and AEMO regions, and supplied bottom levels.

4. **Protection and coherence.** Stored releases are checked using exact nodewise envelope and prefix tests. Aggregation consistency is checked separately within numerical tolerance.

Main tables report three-run means; sample standard deviations (SDs) are given in Supplementary Note S1. For uncertainty analysis, paired percentage changes are averaged over backbones and runs within each dataset, yielding seven dataset-level effects. The retrospective bootstrap, sign-flip tests and Holm correction families are specified in Supplementary Note S4.

6 Results

The seven-dataset tests compare complete forecasting procedures and release rules that satisfy the same protection limits. We report overall, regional and system errors, together with the accuracy cost of protection. The AEMO whole-month test extends the evaluation beyond the short windows.

6.1 Complete-procedure accuracy

Table 6: Window-protocol accuracy.

		Baseline outputs					Protected procedures		
Dataset	Backbone	Raw	NNLS	ProCAST	Ada-Y	PIR	Protected PIR	CPOR-Cov	CPOR-ML
RH	PatchTST	1154.51	1154.15	1080.38	1307.15	**1076.05**	1116.47	948.66	**947.27**
	DLinear	1627.74	1624.06	1459.30	1901.25	**1238.61**	1421.91	**1192.97**	1193.89
	DUET	1265.70	1263.50	1142.72	1529.48	**947.28**	1089.05	953.06	**948.90**
ECommerce	PatchTST	**1498.57**	**1498.57**	1509.91	1500.53	1582.66	1498.49	1476.19	**1470.64**
	DLinear	1546.72	1546.72	1562.45	**1541.79**	1617.31	1537.69	1494.24	**1484.68**
	DUET	1506.43	1506.43	**1505.86**	1513.19	1598.01	1508.94	1484.18	**1479.74**
Traffic	PatchTST	3.824	3.827	5.296	3.841	**3.748**	3.749	**3.739**	3.743
	DLinear	4.118	4.116	5.365	4.131	**3.864**	3.988	**3.977**	3.981
	DUET	4.286	4.291	5.389	4.316	**3.627**	**4.038**	4.117	4.136
Wiki2	PatchTST	**2539.72**	2553.86	3172.20	2546.76	2772.26	2512.58	**2465.83**	2483.48
	DLinear	**2704.95**	2705.86	3363.83	2708.30	2899.81	2643.44	**2548.73**	2557.01
	DUET	**2544.16**	2559.41	3187.99	2551.86	2799.51	2522.99	**2465.16**	2484.33
Labour	PatchTST	15.92	15.86	23.43	21.07	**15.19**	15.36	**14.81**	**14.81**
	DLinear	19.52	19.51	25.61	23.58	**18.57**	18.75	**17.68**	17.69
	DUET	16.39	16.62	22.73	23.38	**12.77**	15.32	15.03	**14.98**
M5	PatchTST	429.73	430.09	429.56	437.70	**417.04**	**417.60**	421.62	420.12
	DLinear	459.08	459.09	465.32	468.52	**427.30**	**440.85**	447.25	445.95
	DUET	457.18	457.35	452.22	442.05	**395.39**	**428.49**	443.35	442.25
AEMO	PatchTST	710.48	707.62	647.72	668.35	**474.21**	654.12	**628.19**	629.02
	DLinear	1227.82	1227.82	1177.44	1181.48	**646.44**	866.66	806.56	**806.52**
	DUET	790.32	785.52	730.73	750.55	**469.56**	655.00	**635.85**	636.05

Three-run mean RMSE after averaging the two windows within each run. Protected PIR applies CPOR full projection to the reconciled PIR target. Boldface marks the lowest displayed RMSE separately within baseline outputs and protected procedures, including ties.

Tables 6 and 7 compare complete procedures in the window and continuous protocols, respectively. Cov and ML reduce RMSE relative to adapted ProCAST by 17.13% and 17.16%, with positive paired reductions in all 21 dataset–backbone groups. The reductions relative to NNLS are 9.32% and 9.33%. These gains compare complete procedures, including how they generate, combine and protect their forecasts.

In the window tests, PIR has the lowest RMSE among baseline outputs on M5 and AEMO for every backbone. Cov and ML improve on PIR in 11 and 10 of the 21 groups. Averaging the paired percentage changes instead gives RMSE increases of 3.21% and 3.19% for Cov and ML. On DUET, their mean error increases are 8.67% and 8.65%; AEMO's short windows, Labour and M5 contribute to these reversals. PIR has no nodewise error constraint in this comparison.

Table 7: Accuracy over 144 continuous releases.

		Baseline outputs			Protected procedures		
Dataset	Backbone	NNLS	ProCAST	PIR	Protected PIR	CPOR-Cov	CPOR-ML
RH	PatchTST	1229.25	**1147.15**	1187.81	1187.20	944.70	**942.76**
	DLinear	1730.64	1543.30	**1342.15**	1464.08	**1111.47**	1115.17
	DUET	1347.18	1222.17	**992.27**	1130.04	911.36	**910.77**
ECommerce	PatchTST	**1611.40**	1621.20	1695.95	1610.79	1580.75	**1574.55**
	DLinear	**1661.30**	1676.50	1731.23	1649.37	1594.12	**1586.34**
	DUET	1619.16	**1617.80**	1713.47	1623.08	1588.26	**1582.03**
Traffic	PatchTST	3.897	5.398	**3.819**	**3.769**	3.774	3.779
	DLinear	4.181	5.464	**3.925**	3.973	**3.969**	3.974
	DUET	4.355	5.483	**3.695**	**3.974**	4.105	4.121
Wiki2	PatchTST	**3116.76**	3734.39	3422.72	3054.27	**2938.79**	2979.91
	DLinear	**3294.74**	3988.23	3572.01	3179.33	**3026.51**	3077.44
	DUET	**3126.86**	3759.16	3458.26	3067.23	**2935.18**	2968.11
Labour	PatchTST	16.79	24.55	**15.56**	16.08	**15.17**	15.18
	DLinear	19.97	26.18	**18.76**	18.95	**16.90**	**16.90**
	DUET	17.46	23.75	**13.37**	15.42	14.90	**14.81**
M5	PatchTST	436.63	436.31	**421.49**	**417.60**	425.75	423.21
	DLinear	467.78	474.85	**433.42**	**440.90**	451.99	450.34
	DUET	463.35	458.64	**400.87**	**417.00**	442.69	441.68
AEMO	PatchTST	709.31	649.84	**476.70**	623.23	**582.29**	582.99
	DLinear	1252.12	1194.62	**651.77**	802.87	667.00	**666.93**
	DUET	787.34	732.08	**473.61**	591.43	556.12	**555.46**

Three-run mean RMSE over all 144 releases without an allowance reset. Protected PIR applies CPOR full projection to the reconciled PIR target. Boldface marks the lowest displayed RMSE separately within baseline outputs and protected procedures, including ties.

In the continuous protocol (Table 7), reductions relative to adapted ProCAST are 20.49% and 20.44%. Cov and ML also improve on PIR by 2.04% and 1.98% overall, although their DUET implementations remain worse by 2.90% and 2.89%. The two protocols use different histories to form targets, different variance weights and different allowance resets. They also compute RMSE over different intervals.

Among the protected procedures in Tables 6–7, Cov and ML reduce RMSE relative to Protected PIR by 3.57% and 3.58% in windows, and 5.67% and 5.59% continuously. Both improve in 17 of 21 window groups; the continuous counts are 16 for Cov and 15 for ML. M5 favors Protected PIR on all three backbones, and Traffic has mixed results. These comparisons use the same protection rule but different target forecasts.

6.2 Accuracy at equal protection

Table 8 fixes the coherent candidate and compares three publication rules under the same protection requirement. Reference, weights, tolerance and feedback are shared; each rule updates its own allowance from its actual releases. With PIR input, direct release means Coherent PIR and full projection means Protected PIR. The table compares each protected rule with its unprotected candidate. The following paired reductions compare one protected rule with another.

For Cov and ML, full projection improves on common shrinkage by 5.17% in windows and by 6.94% and 7.00%, respectively, continuously. All 21 window group means improve; one continuous ML group slightly favors common shrinkage. Full projection also improves on two-group shrinkage on average in both protocols.

We repeat the comparison using the same reconciled forecasts from PIR. Full projection reduces RMSE relative to common shrinkage by 3.57% in windows and 4.34% continuously; reductions relative to two-group shrinkage are 2.72% and 3.28%. Against either shrinkage rule,

Table 8: Protection cost for identical coherent candidates.

		RMSE increase vs. direct release (%)		
Protocol	Candidate	Common shrinkage	Two-group shrinkage	Full projection
Window	Cov	68.19	63.78	**55.66**
Window	ML	65.72	61.38	**53.07**
Window	Coherent PIR	11.39	10.10	**6.82**
Continuous	Cov	60.89	55.27	**45.31**
Continuous	ML	60.20	54.43	**43.95**
Continuous	Coherent PIR	9.32	7.81	**4.01**

Direct release of each coherent candidate is the zero-cost baseline. Percentages are computed within runs and averaged equally over three runs and 21 dataset–backbone groups. Boldface marks the lowest cost among the three protected rules. Full projection of Coherent PIR is Protected PIR.

19 of 21 group means improve in each protocol. Both reversals occur on ECommerce. For the common-shrinkage comparison, the seven-dataset bootstrap 95% intervals are [1.77, 5.96]% and [2.23, 7.07]%, respectively. Supplementary Note S4 reports the retrospective uncertainty analysis for both contrasts.

Common shrinkage lets one restricted node limit corrections throughout the system. In a Labour example with identical pre-release allowances, one South Australian series limits the common step to 0.295% of the target move. Of Western Australia's desired correction of 4.552, common shrinkage retains 0.013 and full projection retains 0.073. This illustrates how full projection preserves more of a candidate correction; its error depends on the subsequently observed value.

6.3 Compliance and protection cost

Table 9: Prefix compliance.

Method	Window ever (%)	Prefixes (%)	Continuous ever (%)	Prefixes (%)
CPOR-Cov	**0.00**	**0.00**	**0.00**	**0.00**
CPOR-ML	**0.00**	**0.00**	**0.00**	**0.00**
Protected PIR	**0.00**	**0.00**	**0.00**	**0.00**
Common shrinkage (PIR)	**0.00**	**0.00**	**0.00**	**0.00**
Two-group shrinkage (PIR)	**0.00**	**0.00**	**0.00**	**0.00**
NNLS reference	**0.00**	**0.00**	**0.00**	**0.00**
Unprotected Cov	80.19	31.87	84.00	28.21
Unprotected ML	79.01	29.54	82.33	24.85
Coherent PIR	80.09	35.84	86.10	36.59
PIR	78.16	32.24	84.17	32.74
Adapted ProCAST	95.56	79.05	96.73	80.64
Ada-Y	85.88	46.32	89.76	45.76

Ever is the percentage of node sequences with at least one cumulative violation; Prefixes is the percentage of violating cumulative-prefix checks. Windows contain 9,612 node sequences; continuous tests contain 4,806. Each protocol has 692,064 node–date checks per method.

Table 9 shows that unprotected targets frequently exceed individual cumulative limits. All protected paths pass the exact nodewise envelope and cumulative-prefix checks, with coherence verified at the specified floating-point tolerances.

Figure 2 shows the ever-violation proportions before protection and the accuracy cost of protecting the same Cov, ML or Coherent PIR candidate. All protected proportions are zero.

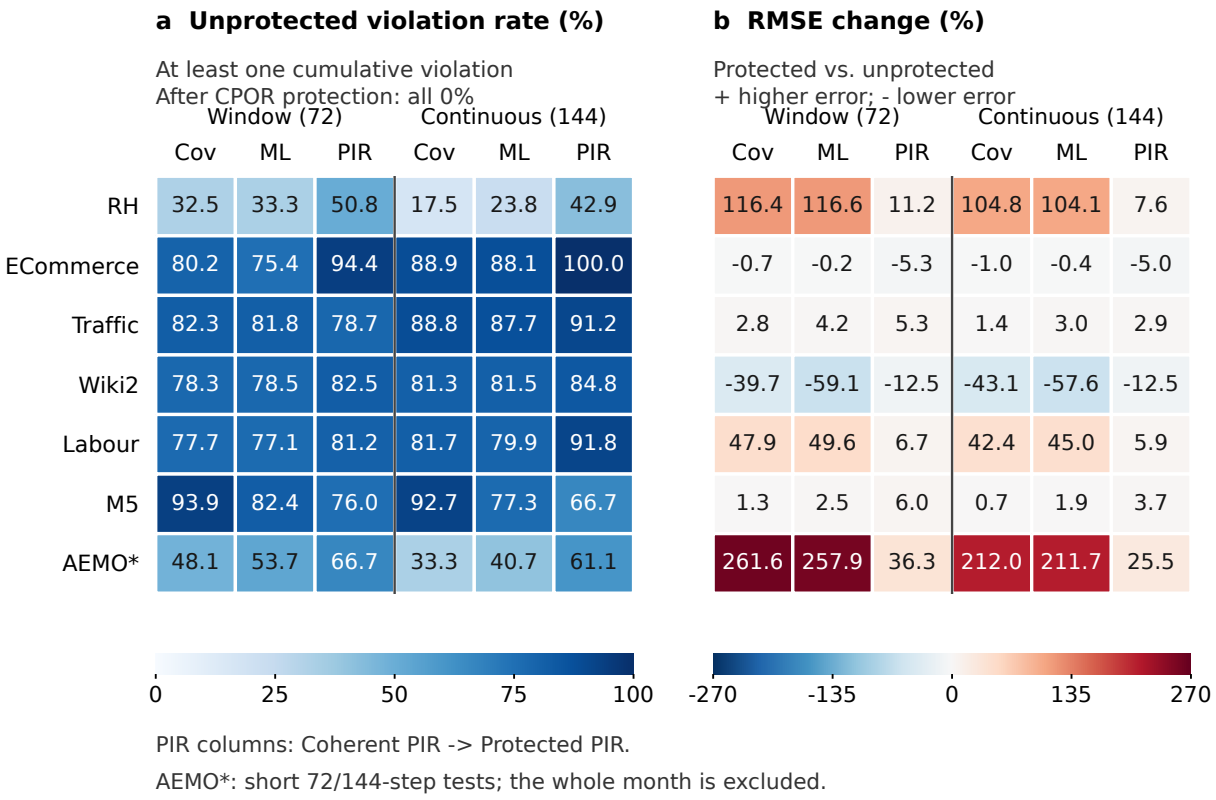


Figure 2: Protection compliance and accuracy cost. (a) Percentage of unprotected node sequences that violate the cumulative limit at least once; protected values are all 0%. (b) Mean-node RMSE change after protecting the same coherent candidate; positive means higher error. Costs average paired percentages over three backbones and three runs. PIR columns compare Coherent PIR with Protected PIR, alongside the corresponding Cov and ML paths. AEMO* uses the short 72/144-step tests, excluding the whole month.

A few large error increases raise the mean protection cost well above the median. Across 63 matched run comparisons, median Cov/ML costs are 3.68%/5.10% in windows and 2.00%/3.46% continuously, well below the means in Table 8. AEMO's short windows, RH and Labour have large costs, whereas Wiki2 and ECommerce improve after protection on average. For Traffic/Cov, protection removes an 82.3% window ever-violation rate at a 2.8% RMSE cost. Frequent violations can therefore coexist with a small accuracy cost. AEMO's larger costs concern its short tests, not the whole-month evaluation.

Protected PIR increases RMSE by 6.82% in windows and 4.01% continuously relative to Coherent PIR. By comparison, its RMSE is 3.57% and 4.34% lower than common shrinkage under the same protection limits. These costs use each method's own unprotected coherent candidate; Tables 6–7 compare the absolute protected RMSEs.

6.4 Continuous allowances

Retaining allowances reduces 144-observation RMSE by 4.38% for Cov and 4.32% for ML relative to the midpoint-reset control, with improvements in all 21 groups. The DUET reductions are 4.47% and 4.46%. Because the paths share targets, weights and selection histories and differ only in the midpoint allowance reset, these gains quantify the value of carrying unused allowance forward. Paired RMSEs and training-run variation are given in Supplementary Note S3.

The protection applies from the agreed start of the full trajectory. When the second half is assessed as a separate zero-allowance contract, 3,304 of 4,806 Cov node sequences and 3,326 ML

sequences exceed that additional limit at least once. Thus protection from the original start does not imply protection for a window starting halfway through the sequence.

6.5 Target choices and aggregation levels

Table 10 compares target-generation and reconciliation choices under the same protection operator and error contract.

Table 10: Protected controls in the window protocol.

Dataset	Backbone	WLS(var)	MinT	Rolling MinT	MinT(Cov)	CalBest	CPOR-Cov	CPOR-ML
RH	PatchTST	1154.14	1149.76	1145.48	948.45	**946.45**	948.66	947.27
	DLinear	1625.26	1601.91	1584.01	1193.06	1193.96	**1192.97**	1193.89
	DUET	1263.57	1254.01	1245.80	953.08	954.49	953.06	**948.90**
ECommerce	PatchTST	1498.57	1498.57	1498.57	1476.19	1500.72	1476.19	**1470.64**
	DLinear	1546.72	1546.72	1546.72	1494.24	1527.00	1494.24	**1484.68**
	DUET	1506.43	1506.43	1506.43	1484.18	1499.66	1484.18	**1479.74**
Traffic	PatchTST	3.826	3.823	3.820	**3.739**	3.783	**3.739**	3.743
	DLinear	4.118	4.116	4.118	**3.970**	4.038	3.977	3.981
	DUET	4.283	4.284	4.284	**4.084**	4.163	4.117	4.136
Wiki2	PatchTST	2541.15	2538.09	2533.80	2475.96	2503.09	**2465.83**	2483.48
	DLinear	2705.07	2703.44	2704.65	2556.28	2597.22	**2548.73**	2557.01
	DUET	2545.24	2541.08	2540.83	2476.49	2503.36	**2465.16**	2484.33
Labour	PatchTST	15.80	15.78	15.83	**14.80**	14.85	14.81	14.81
	DLinear	19.52	19.51	19.51	**17.68**	17.85	**17.68**	17.69
	DUET	16.26	16.11	15.72	15.06	15.01	15.03	**14.98**
M5	PatchTST	429.45	429.45	429.50	421.35	429.47	421.62	**420.12**
	DLinear	459.06	459.06	459.06	447.03	459.06	447.25	**445.95**
	DUET	456.88	456.58	456.53	443.25	456.87	443.35	**442.25**
AEMO	PatchTST	707.24	705.29	704.08	**627.56**	628.39	628.19	629.02
	DLinear	1227.82	1227.82	1227.82	806.57	**806.51**	806.56	806.52
	DUET	783.18	780.26	775.52	635.81	**635.27**	635.85	636.05

Three-run mean RMSE. WLS and MinT use raw forecasts unless Cov input is stated. CalBest fixes the expert selected during calibration. All listed controls use CPOR publication and their own allowances.

Cov and ML reduce window RMSE relative to raw-input protected MinT by 8.94% and 8.95%. When MinT uses the Cov target, the window gains fall to approximately zero for Cov and 0.02% for ML. The continuous gains are 0.17% and 0.08%. The smaller differences show that the target forecast contributes to the raw-input comparison.

Window gains over protected CalBest are 1.04% and 1.06%, comparing adaptive expert combinations with one fixed expert.

Regional and system errors. Table 11 reports selected system and regional levels in original units. NNLS provides the protection reference; Protected PIR, Cov and ML share the CPOR protection operator but construct different targets. Across the six datasets with observed system totals, Cov and ML reduce system RMSE relative to NNLS by 10.26% and 10.14% in windows, and 14.58% and 14.35% continuously.

Against Protected PIR, Cov and ML reduce system RMSE by 3.17% and 3.05% in windows, and 5.17% and 4.93% continuously, averaging paired reductions over the six datasets, three backbones and three runs. For Labour and AEMO regional means, the corresponding reductions are 4.38%/4.42% and 8.50%/8.57%; all six dataset–backbone regional groups favor Cov and ML in both protocols. The protection rule is the same; the methods form different targets.

Lower all-node RMSE does not imply lower system-total RMSE. Protected PIR outperforms both implementations on Traffic with every backbone and on M5 with every backbone in continuous tests. For Traffic/DUET, continuous system RMSE increases by 8.51% and 8.92% relative

Table 11: System totals and regional means.

Level	Backbone	Window RMSE				Continuous RMSE			
		NNLS	Protected PIR	Cov	ML	NNLS	Protected PIR	Cov	ML
RH total	PatchTST	1854.50	1790.93	1539.80	**1536.06**	1972.54	1908.37	1540.11	**1535.33**
	DLinear	2699.48	2360.63	**2008.90**	2013.50	2870.68	2424.68	**1867.63**	1876.15
	DUET	2063.65	1767.52	1559.71	**1553.54**	2196.57	1841.98	1493.37	**1493.25**
Traffic total	PatchTST	129.084	**127.887**	129.181	129.280	129.107	**126.451**	129.328	129.406
	DLinear	135.697	**134.088**	135.818	135.913	135.988	**132.126**	136.323	136.376
	DUET	157.682	**146.998**	154.454	155.212	157.739	**141.343**	153.361	153.952
Wiki2 total	PatchTST	45757.52	44838.56	**43695.71**	44235.02	56113.77	54548.89	**52096.53**	52914.19
	DLinear	50854.17	48947.47	**47490.32**	47672.40	60258.95	57196.79	**55115.64**	55424.34
	DUET	46635.81	45657.24	**44476.20**	44924.96	56961.89	55304.44	**52171.74**	53079.65
Labour total	PatchTST	146.65	141.32	**138.54**	139.06	161.37	152.97	**148.96**	149.94
	DLinear	197.96	190.79	**178.80**	178.90	202.12	192.79	**172.45**	172.79
	DUET	172.16	153.92	152.75	**152.00**	185.12	156.42	153.59	**152.32**
M5 total	PatchTST	3835.10	3725.18	3734.70	**3720.15**	3915.04	**3741.14**	3774.31	3761.04
	DLinear	4141.00	**3978.64**	4013.36	4002.33	4264.78	**4025.71**	4088.14	4085.04
	DUET	4175.00	**3883.28**	4014.92	4009.19	4253.76	**3772.77**	4001.35	4006.67
AEMO total	PatchTST	1974.54	1813.51	**1770.21**	1775.21	1977.24	1713.76	**1637.48**	1643.53
	DLinear	3634.86	2545.96	2376.70	**2376.56**	3705.81	2353.63	1963.76	**1963.52**
	DUET	2267.36	1879.09	**1833.23**	1834.28	2268.45	1670.94	1589.44	**1588.29**
Labour regions	PatchTST	23.08	22.34	**21.71**	21.74	24.99	23.86	**22.90**	22.99
	DLinear	30.62	29.43	**27.67**	27.68	31.56	29.99	**26.57**	26.60
	DUET	25.61	23.31	22.95	**22.86**	27.43	23.76	23.05	**22.91**
AEMO regions	PatchTST	454.23	422.24	399.79	**399.78**	455.73	405.13	371.26	**370.88**
	DLinear	746.41	530.80	492.53	**492.51**	761.38	492.72	407.65	**407.61**
	DUET	489.15	410.18	**396.37**	396.41	491.12	375.53	349.46	**348.89**

Three-run mean RMSE in original units; Cov and ML denote CPOR-Cov and CPOR-ML. Boldface marks the lowest displayed error within each protocol and row, including ties. Protected PIR uses the same reference rule, tolerance and CPOR full projection, and updates its own allowance from its releases. Bottom-level results and training-run SDs are given in Supplementary Note S3.1.

to Protected PIR. Traffic/DLinear also retains the smaller reversals against NNLS: 0.09%/0.16% in windows and 0.25%/0.29% continuously, despite lower all-node RMSE.

6.6 Longer continuous demand evaluation

Table 12 reports whole-month accuracy for all three backbones, with one continuous allowance trajectory per method and run.

Under the same CPOR protection rule, Cov and ML reduce whole-month mean-node RMSE relative to Protected PIR by 51.99% and 52.59%, averaging paired percentages over the nine backbone–run streams. Regional-mean reductions are 53.78% and 54.41%, and system reductions are 49.38% and 49.94%. Cov and ML improve in all nine backbone–run comparisons at each reported level. These methods construct different targets and use the same protection rule.

On identical Cov and ML targets, full projection improves on common shrinkage by 8.88% and 8.91%, averaged over the nine backbone–run streams. Relative to direct release, its costs are 41.38% and 43.06%. All protected month-long paths satisfy the limits. The first comparison measures which protected rule retains more accuracy; the second measures the error increase caused by requiring protection.

ML nearly matches the calibration-selected fixed expert on DLinear. With PatchTST, Cov and ML increase QLD RMSE relative to common shrinkage by 3.74% and 3.46%, averaged over three runs. All six node results appear in Supplementary Note S5.

Table 12: AEMO whole-month accuracy.

Level	Method	PatchTST	DLinear	DUET
Overall	NNLS reference	667.19 ± 29.72	1234.14 ± 17.39	757.24 ± 16.46
	Adapted ProCAST	614.34 ± 29.16	1174.72 ± 14.22	698.82 ± 12.89
	Ada-Y	672.74 ± 34.59	1198.84 ± 14.54	743.40 ± 18.94
	PIR	609.65 ± 26.68	709.37 ± 4.78	480.91 ± 4.97
	Unprotected Cov	190.23 ± 0.48	198.40 ± 0.47	200.82 ± 0.58
	Unprotected ML	183.22 ± 0.63	197.22 ± 0.00	195.53 ± 0.93
	Protected PIR	569.58 ± 30.11	719.98 ± 4.08	477.74 ± 6.06
	Protected MinT (Cov)	293.54 ± 12.11	288.62 ± 2.54	252.15 ± 2.33
	Protected fixed expert	425.74 ± 21.32	$\mathbf{287.82 \pm 2.23}$	352.84 ± 91.79
	CPOR-Cov	291.10 ± 13.19	288.59 ± 2.53	252.37 ± 2.33
	CPOR-ML	$\mathbf{286.47 \pm 13.27}$	$\mathbf{287.82 \pm 2.23}$	$\mathbf{248.10 \pm 2.04}$
Regions	NNLS reference	459.66 ± 14.86	753.70 ± 10.94	484.16 ± 8.75
	Protected PIR	414.78 ± 13.48	448.19 ± 1.90	332.28 ± 3.99
	CPOR-Cov	202.70 ± 6.39	179.78 ± 1.51	165.09 ± 2.06
	CPOR-ML	$\mathbf{199.07 \pm 6.47}$	$\mathbf{179.26 \pm 1.30}$	$\mathbf{162.03 \pm 1.83}$
System	NNLS reference	1704.85 ± 104.71	3636.33 ± 49.70	2122.64 ± 55.12
	Protected PIR	1343.59 ± 114.25	2078.95 ± 15.44	1205.00 ± 25.66
	CPOR-Cov	733.06 ± 47.34	832.63 ± 7.65	688.80 ± 6.54
	CPOR-ML	$\mathbf{723.47 \pm 47.59}$	$\mathbf{830.59 \pm 6.86}$	$\mathbf{678.44 \pm 6.25}$

RMSE in MW, three-run mean ± SD over 1,488 half-hours without an allowance reset. Overall averages six node RMSEs; Regions averages the five regional RMSEs; System uses the total-demand series. Boldface compares protected procedures within each level and backbone. Protected PIR, Cov and ML share the NNLS reference and CPOR protection rule, with independently updated allowances.

6.7 Supporting analyses and computation

The outcome-domain comparison tests the value of using known nonnegativity. Allowing outcomes to take any real value gives the more restrictive envelope

$$\Gamma_R^{\mathbb{R}}(\beta, x) = \frac{R(x-\beta)^2}{R-1}. \tag{23}$$

On the historical six-dataset, two-backbone window grid, using the nonnegative envelope improves Cov and ML RMSE by 1.970% and 1.958% relative to this control; Labour/DLinear slightly favors the all-real envelope.

We also compare the two-stage procedure with directly projecting the raw combined target a_t:

$$q_t^{\text{direct}} \in \arg\min_{q \in \mathcal{K}_t^{\text{direct}}} \frac{1}{2} \sum_i (q_i - a_{t,i})^2 / v_i. \tag{24}$$

Here $\mathcal{K}_t^{\text{direct}}$ uses the direct path's own remaining allowance. On the historical streams, the two orders have nearly identical overall RMSE, with small system-level reversals. Supplementary Note S6 reports both the domain and projection-order comparisons.

The supplementary recent-method comparisons use fewer datasets or training runs than the main grid. On four eligible datasets with one PatchTST/DLinear training run, Cov and ML achieve lower overall RMSE than both Li preprint configurations in all 16 comparable windows, while Traffic retains system-level reversals. Supplementary Note S8 reports the Li, residual-correction, COSA, robust-map and MCGrad comparisons. Comparisons between different targets measure complete procedures. Comparisons using the same coherent target measure the release rule.

In the dedicated six-dataset, two-backbone timing study, ML is faster in all twelve groups; Cov's mean total postprocessing time is 1.90–4.34 times ML's. Cov estimates a 9×9 error-moment matrix and solves nodewise combination problems, whereas ML updates weights recursively. Both enforce the same protection requirement. Thus ML is the less costly choice in the measured

settings; their accuracy differences depend on the dataset and level (Tables 6 and 11). Detailed timings and numerical events are given in Supplementary Note S7.

7 Conclusion

CPOR makes a nodewise cumulative error contract computable for coherent online forecasts. For nonnegative outcomes with no known upper bound, its intervals characterize the exact one-step protection set. Updating allowances from the errors of actual releases extends protection to every prefix from the agreed start. Cov and ML implement statistical and sequential target selection under this common protection mechanism.

Tests on seven datasets, three backbones and three training runs examine three choices: how to form the target, how to protect it, and whether to retain unused allowance. On identical Coherent PIR candidates, full projection reduces mean-node RMSE relative to common shrinkage by 3.57% in window tests and 4.34% in continuous tests, and also improves on two-group shrinkage on average. Retaining allowances reduces Cov and ML RMSE by 4.38% and 4.32% relative to matched midpoint resets. All tested protected paths satisfy the cumulative limits.

Complete Cov and ML procedures improve on adapted ProCAST and Protected PIR on average. Average paired comparisons favor unprotected PIR in window tests and CPOR in continuous tests. With DUET, PIR is more accurate in both. Protection can increase error, and some regions or system totals deteriorate. The AEMO whole-month test extends the evaluation beyond short windows. Full projection is useful when these cumulative limits are required, retaining more accuracy on average than the tested shrinkage rules. Future work will study delayed feedback and larger aggregation structures.

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Data availability. RH and ECommerce come from the ProCAST repository [25]. Traffic, Wiki2 and Labour use the hierarchical-data archive [26]. M5 uses the retail archive [27]. AEMO demand and rooftop-PV sources are listed in Section 5.1.

Code availability. Code and experimental settings are available at `https://github.com/kavin027/CPOR/tree/scientific-review-20261109`.

Supplementary information. Detailed results and analyses accompany the article.

Use of AI-assisted tools. OpenAI Codex assisted with experimental code development, debugging, result analysis and manuscript preparation. AI tools also assisted the initial design of Figure 1, which was subsequently edited by the author. The author reviewed the material and takes responsibility for the final content.

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